PROJECT PRESENTATION

STUDENT MANAGEMENT SYSTEM

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PROJECT

Source Code

The complete code of the Student Management System is shared as plain text on Pastebin, so anyone can open, read and copy it.

Code link

https://pastebin.com/raw/iw2ZNGSm

PROJECT

Output

The final output of the project is published on Netlify. It opens in any web browser, with no installation needed.

Live website

https://alan-project-studentmanagment.netlify.app/

AWS Services Used

01
VPC
Private network

02
Subnet & Route Table
Traffic routing

03
Internet Gateway
Internet access

04
EC2 & Nginx
Web server

05
S3
Backup storage

06
Lambda
Serverless code

07
IAM
Permissions

08
SNS
Email alerts

09
CloudWatch
Logs and metrics

01 · NETWORKING

VPC

A Virtual Private Cloud (VPC) is your own private network inside AWS.

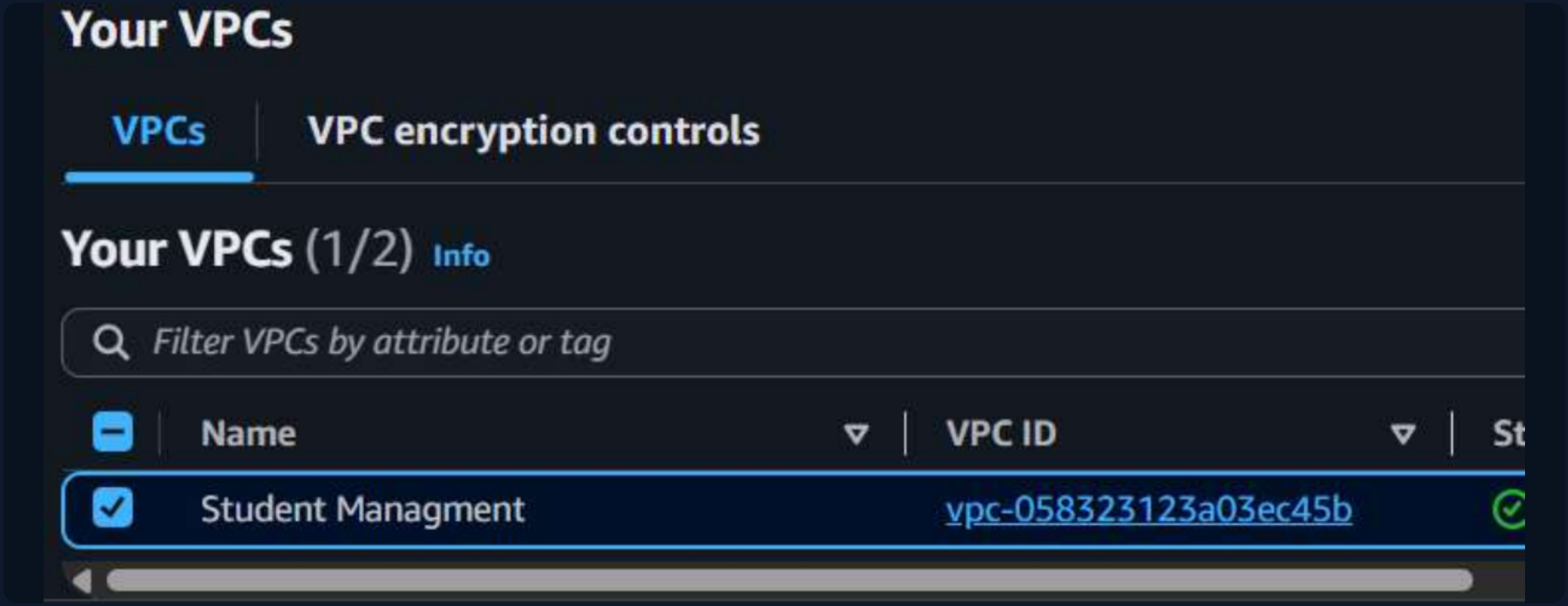
All the resources of this project run inside it, separated from other users.

In this project

- A dedicated VPC for the project
- IP range 10.0.0.0/16
- Created in Europe (Stockholm)

VPC

The project VPC created in AWS



02 · NETWORKING

Subnet & Route Table

A subnet is a smaller section of the VPC where the server is placed.

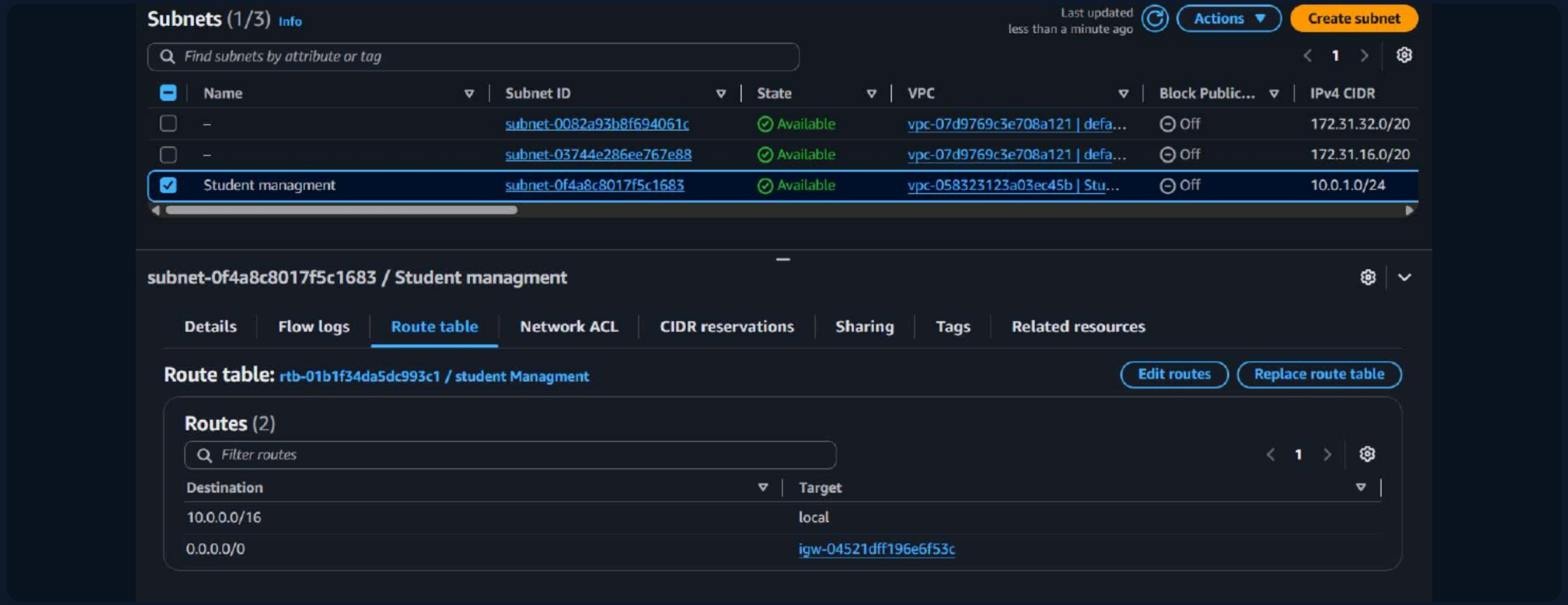
A route table decides where the network traffic is sent.

In this project

- Subnet range 10.0.1.0/24
- Local route for 10.0.0.0/16
- Route 0.0.0.0/0 to the internet gateway

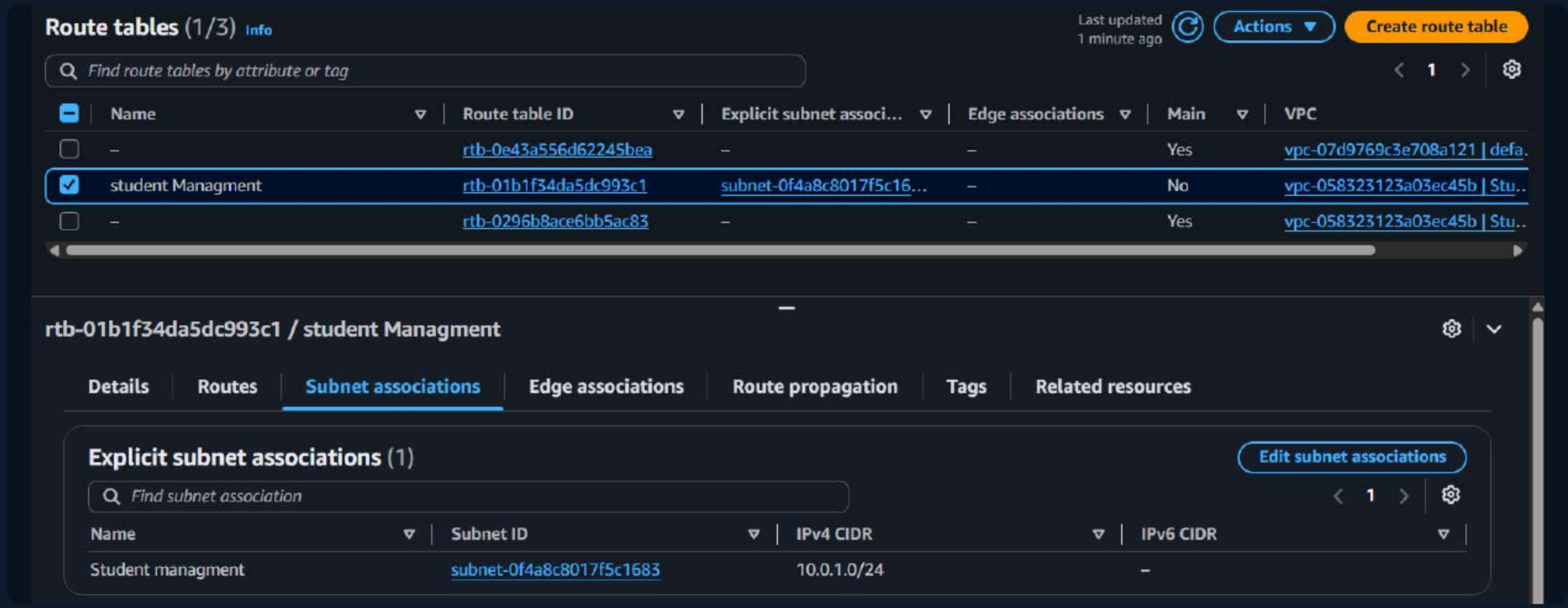
Subnet & Routes

Subnet 10.0.1.0/24 with a local route and an internet route



Route Table Association

The project route table is linked to the project subnet



03 · NETWORKING

Internet Gateway

An Internet Gateway connects the VPC to the internet.

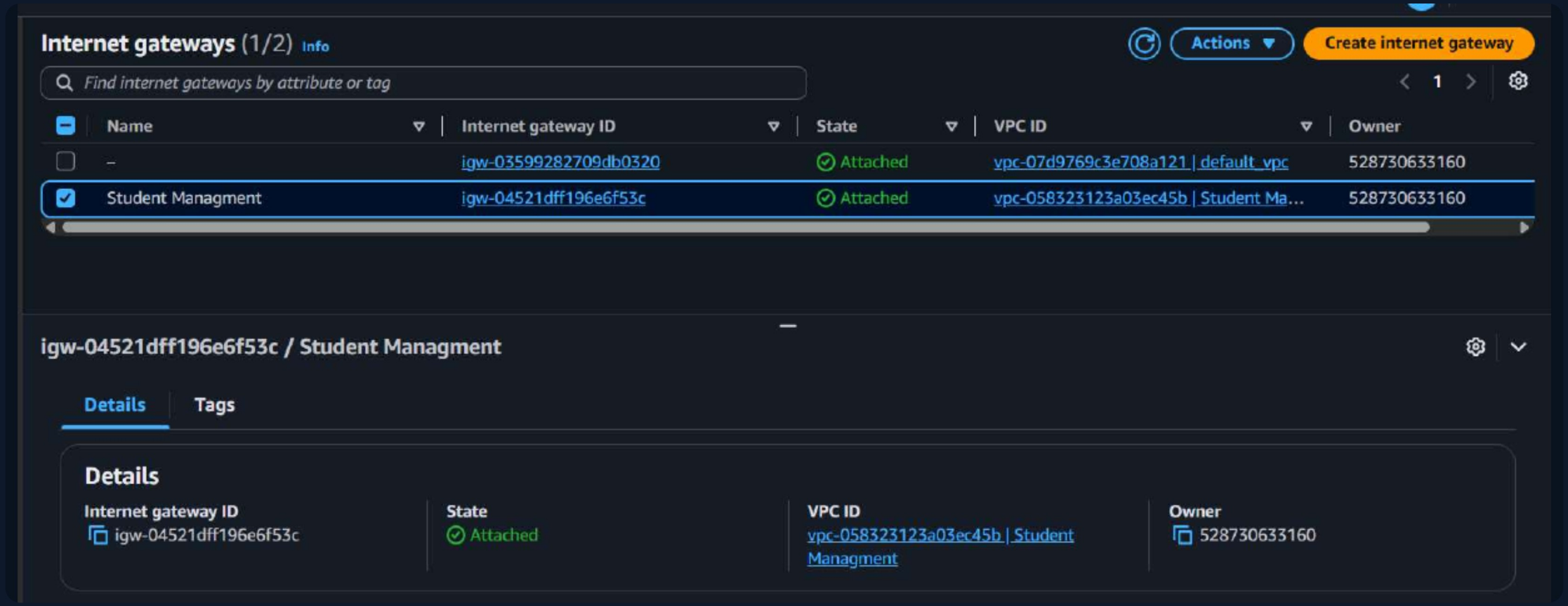
Without it, users could not open the web server from their browsers.

In this project

- Attached to the project VPC
- Route 0.0.0.0/0 points to it
- Makes the subnet public

Internet Gateway

The gateway is attached to the project VPC



04 · COMPUTE

EC2 Instance

Elastic Compute Cloud (EC2) gives virtual servers in AWS.

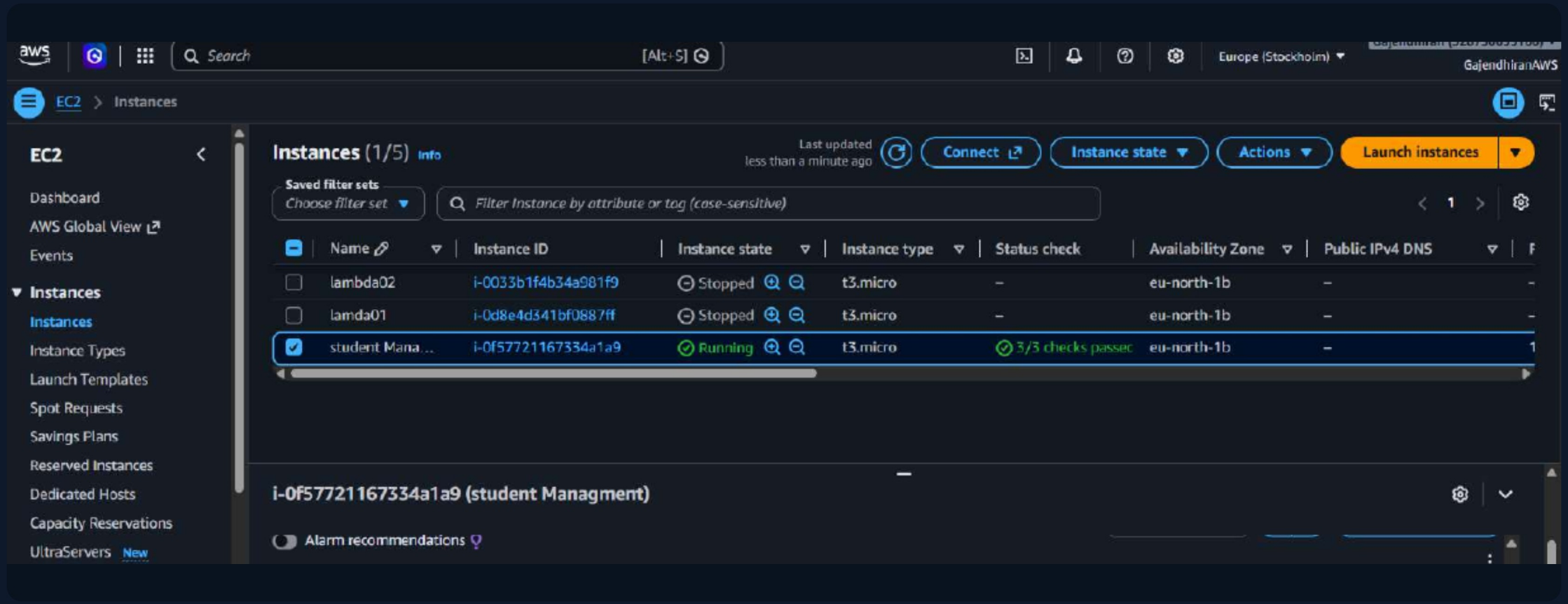
The Student Management web page is hosted on an EC2 instance.

In this project

- Instance type t3.micro
- Amazon Linux 2023
- State Running, 3/3 status checks passed

EC2 Instance

The student management instance is running



05 · COMPUTE

Nginx Web Server

Nginx is a web server that shows the project page to visitors.

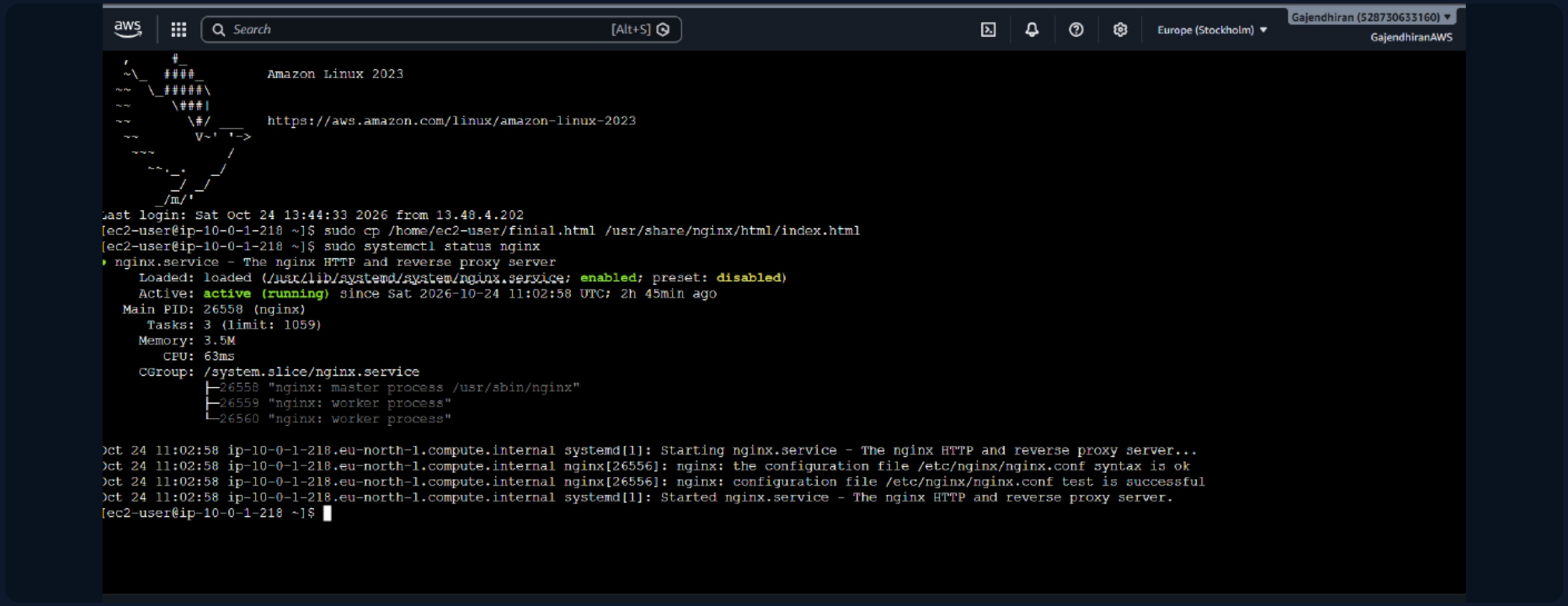
The project HTML file is copied into the Nginx web folder.

In this project

- Page copied to /usr/share/nginx/html/index.html
- Service status: active (running)
- Enabled to start automatically

Nginx Web Server

Page copied to Nginx and the service is active (running)



06 · STORAGE

Amazon S3

Simple Storage Service (S3) stores files as objects inside buckets.

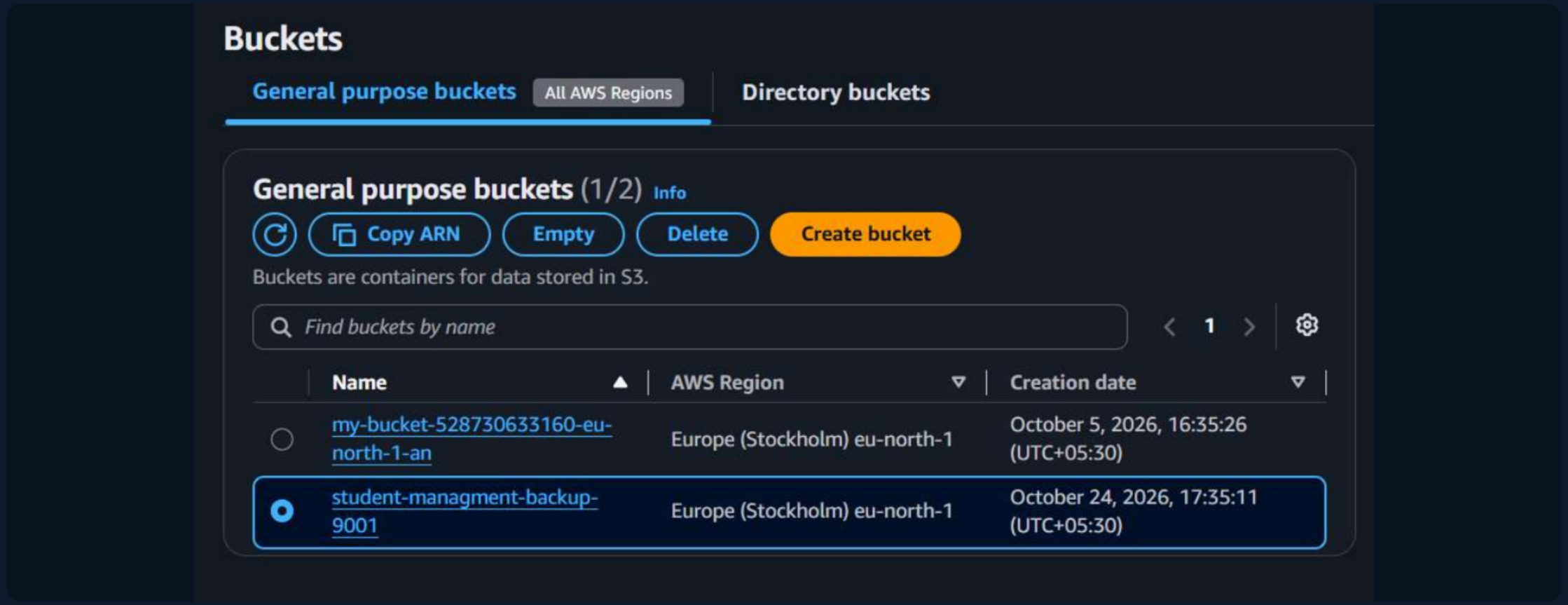
It is used here to keep the backup of the student data.

In this project

- Bucket: student-managment-backup-9001
- Region: Europe (Stockholm)
- Backup file uploaded: std manage.txt

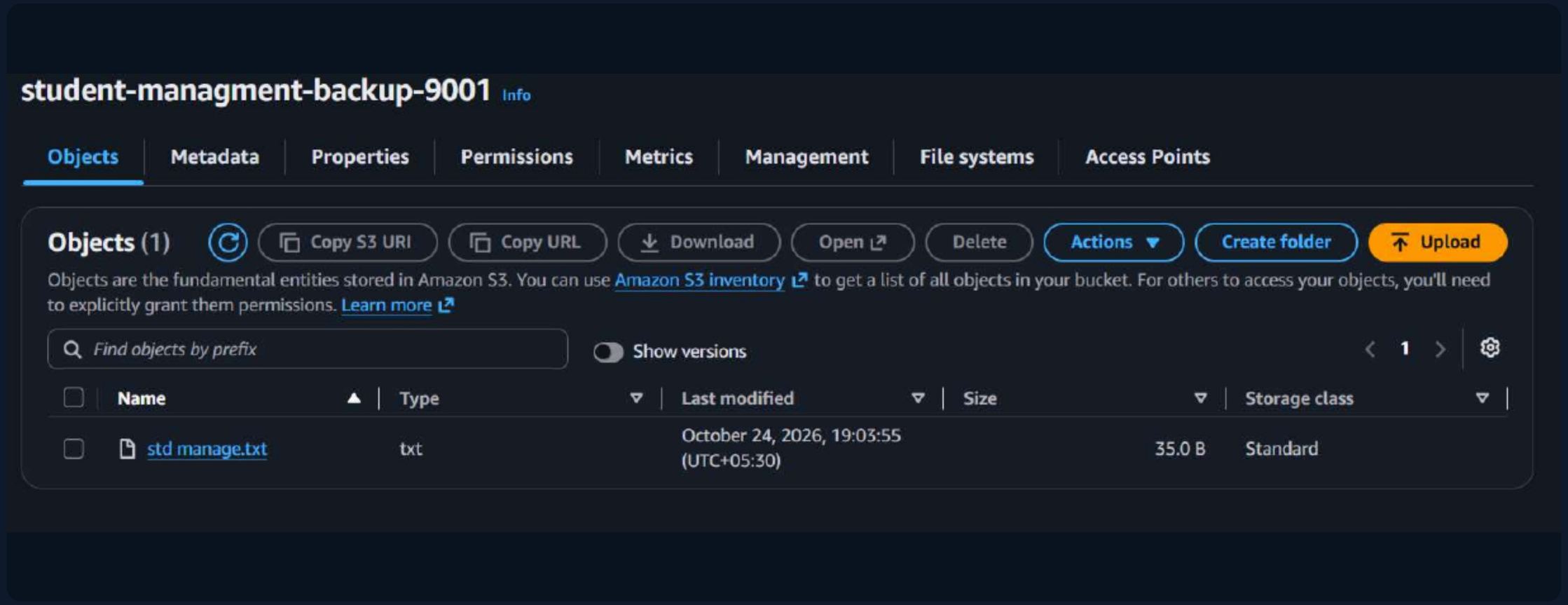
S3 Bucket

The backup bucket created in Europe (Stockholm)



Backup File in S3

The backup file uploaded to the bucket



07 · SERVERLESS

AWS Lambda

Lambda runs code without managing any server.

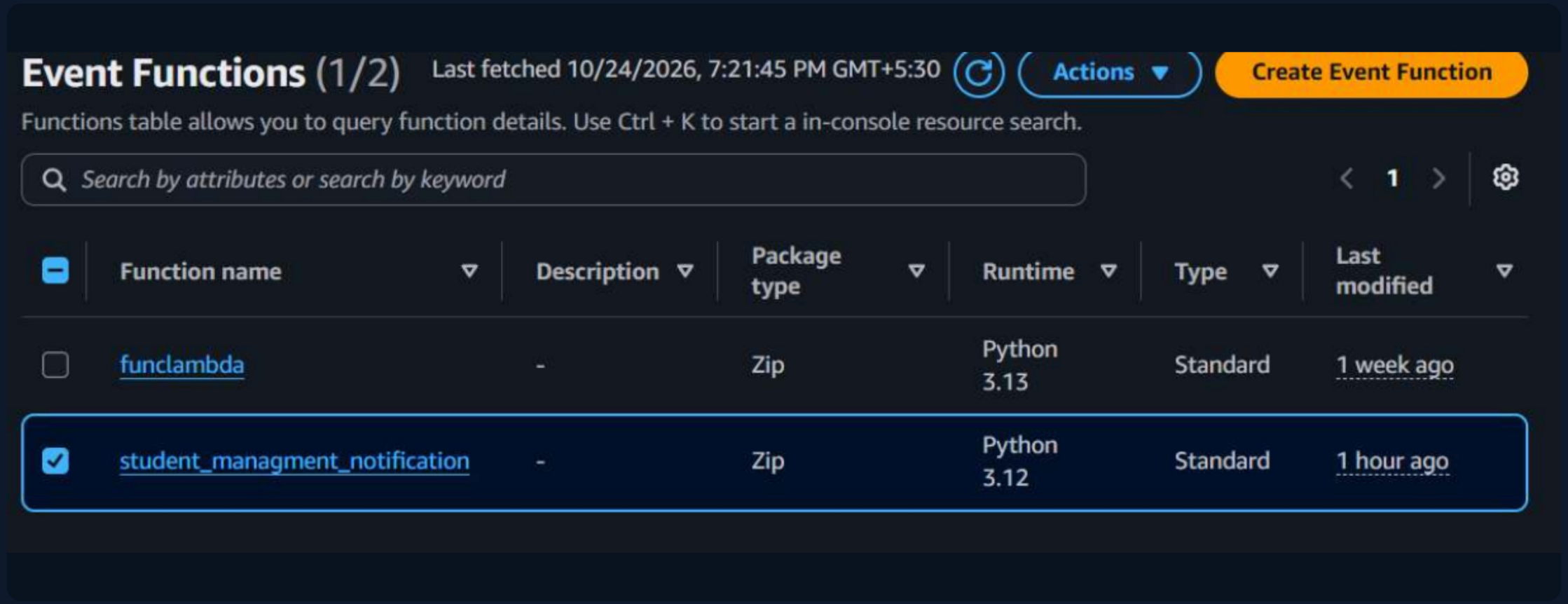
Our function is written to handle S3 upload events and send an email notification through SNS.

In this project

- student_managment_notification, Python 3.12
- Reads the bucket and file name from the event
- Publishes the message to the SNS topic
- Topic set in the TOPIC_ARN variable

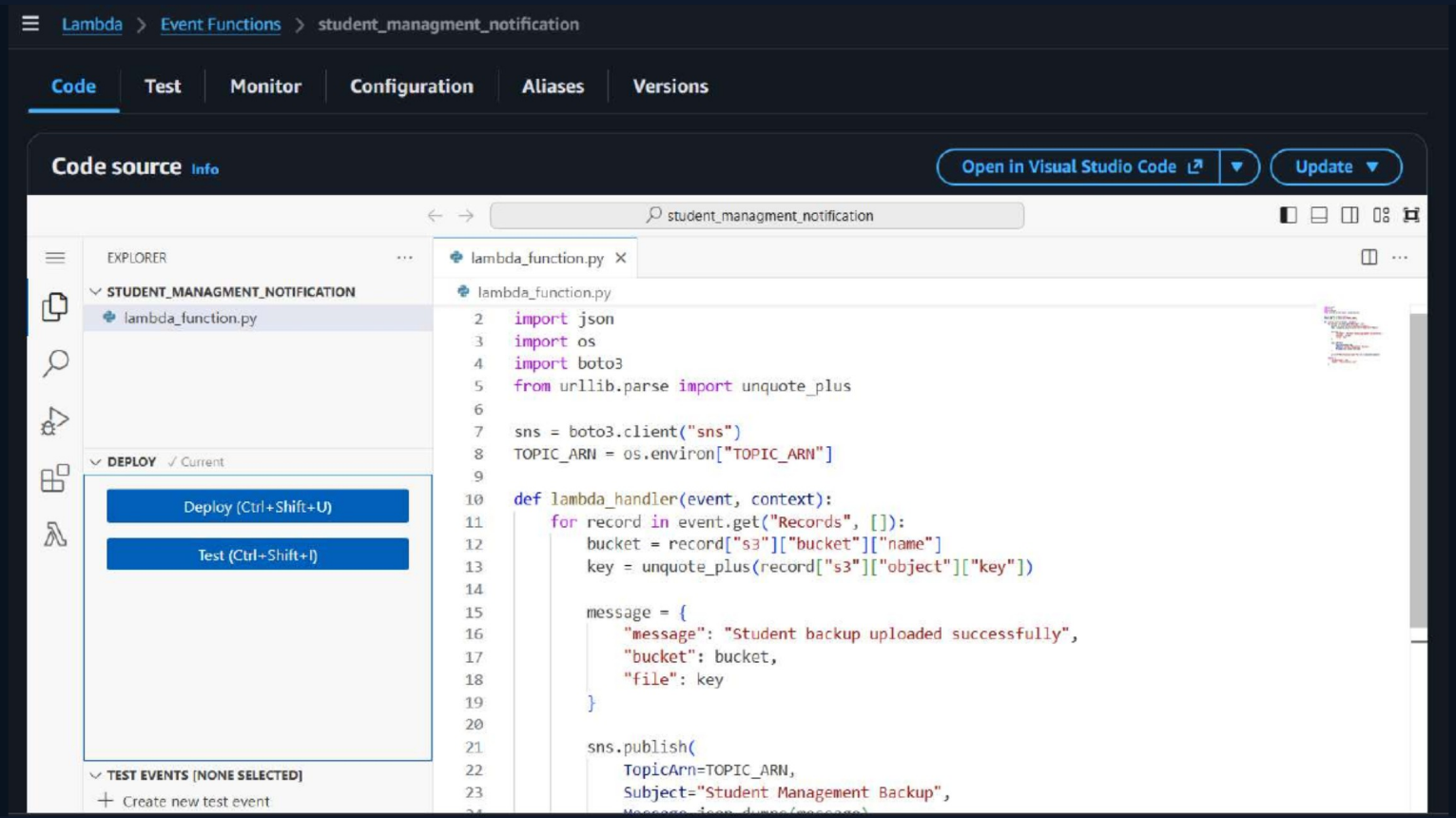
Lambda Function

student_managment_notification, Python 3.12



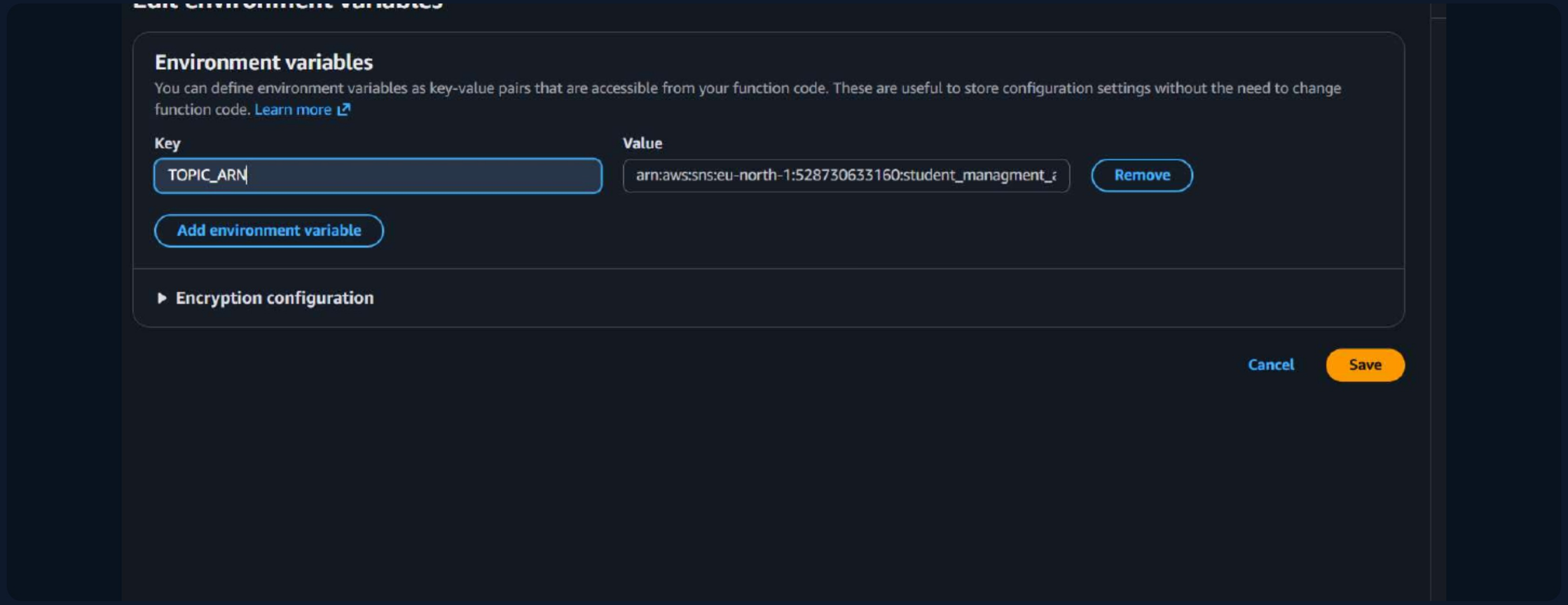
Lambda Code

Reads the S3 event and publishes a message to SNS



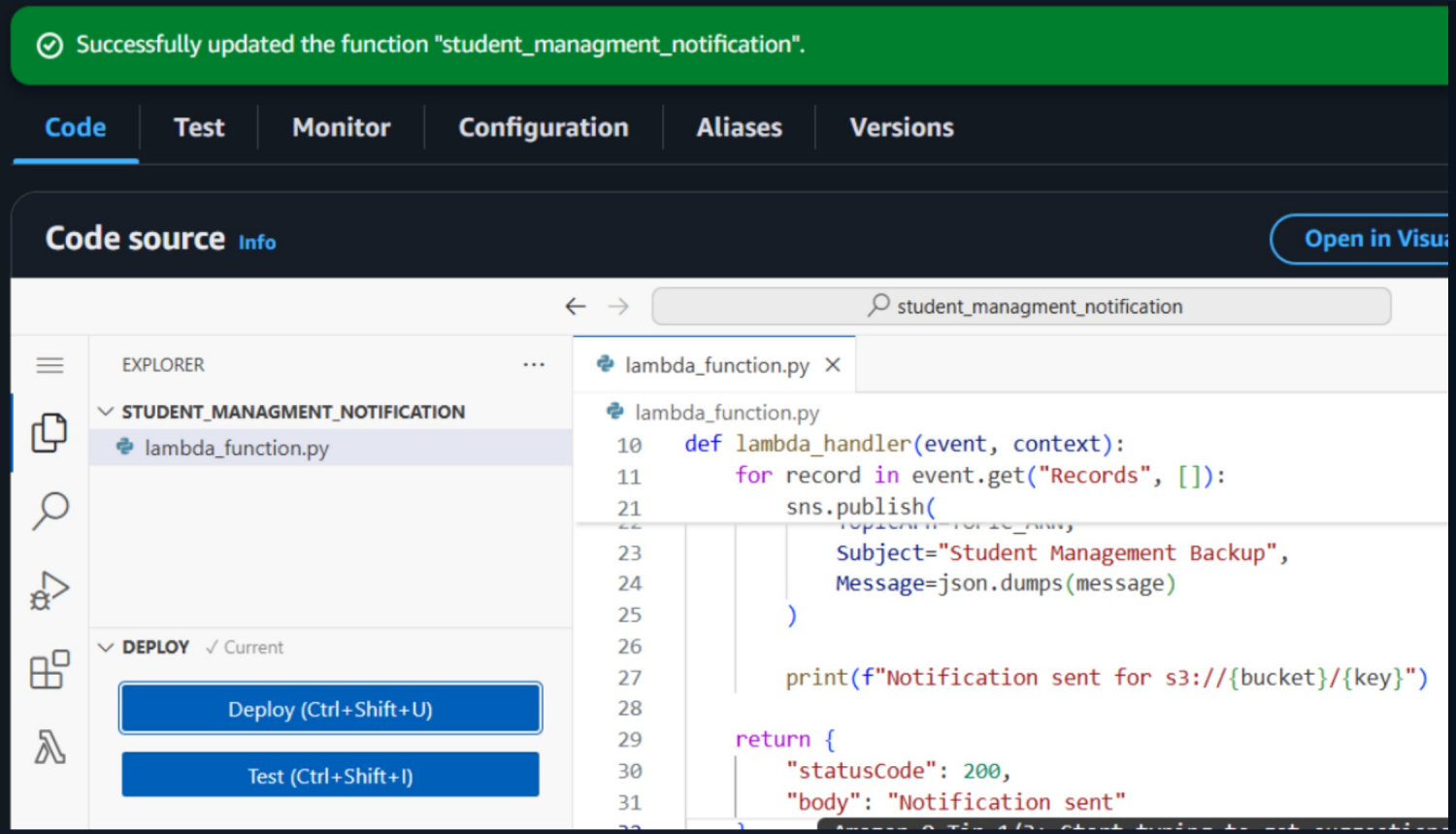
Environment Variable

TOPIC_ARN holds the address of the SNS topic



Function Deployed

The function was updated successfully



08 · SECURITY

IAM Role

Identity and Access Management (IAM) controls who and what can use AWS services.

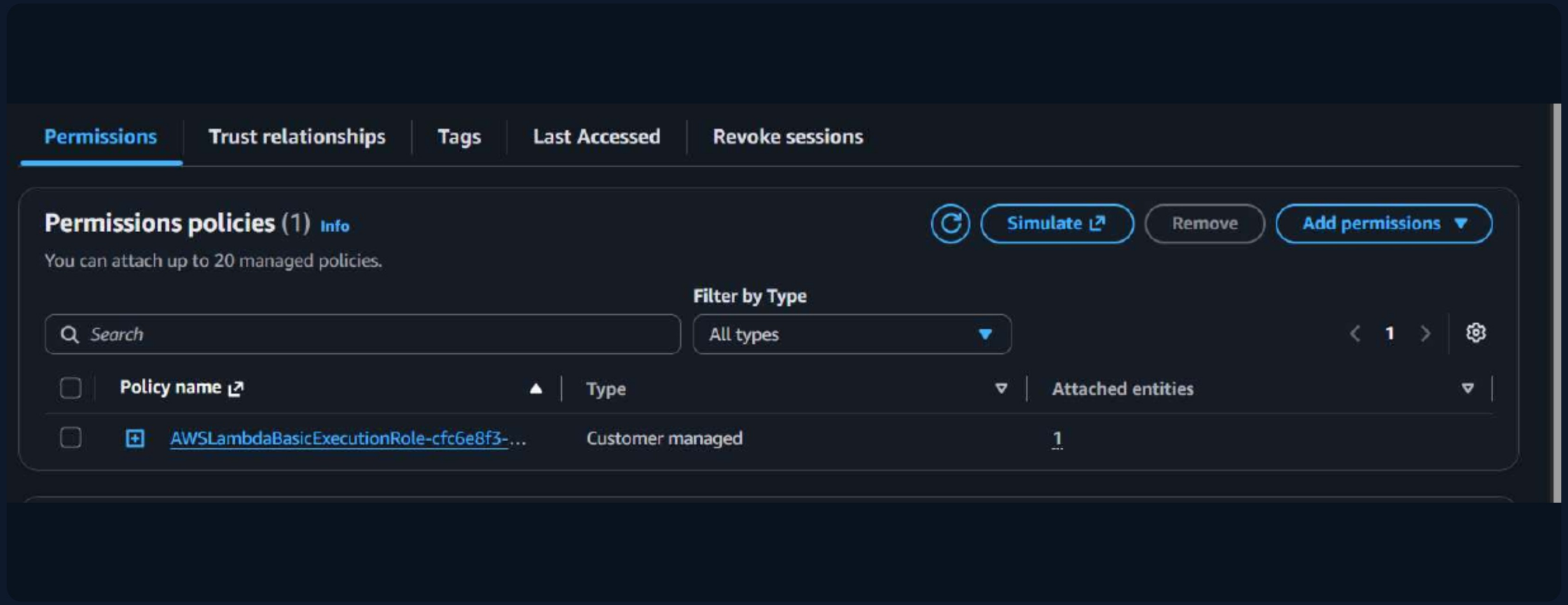
The Lambda function runs with an execution role that holds its permissions.

In this project

- Execution role of the Lambda function
- Policy: AWSLambdaBasicExecutionRole
- Lets the function write logs

IAM Permissions

AWSLambdaBasicExecutionRole policy attached to the role



09 · NOTIFICATIONS

Amazon SNS

Simple Notification Service (SNS) sends messages to everyone subscribed to a topic.

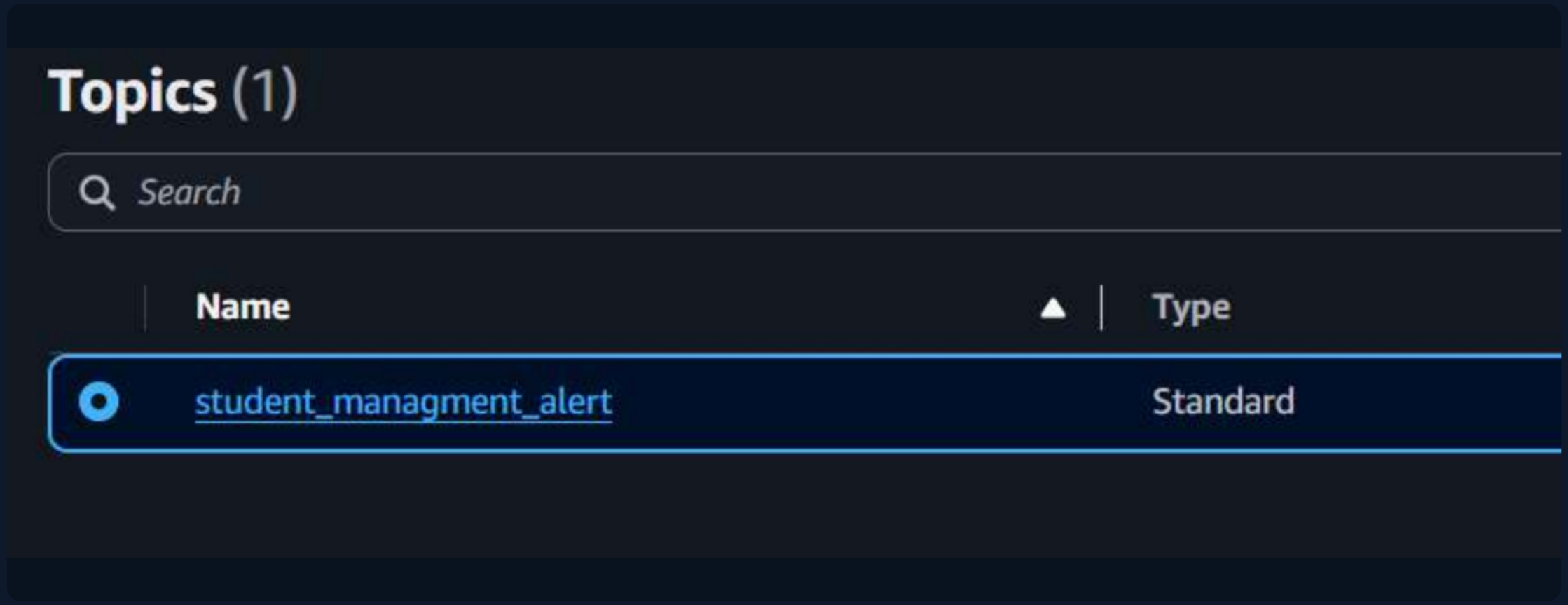
Here it sends an email alert about the student backup.

In this project

- Topic: student_managment_alert
- Protocol: Email
- Subscription status: Confirmed

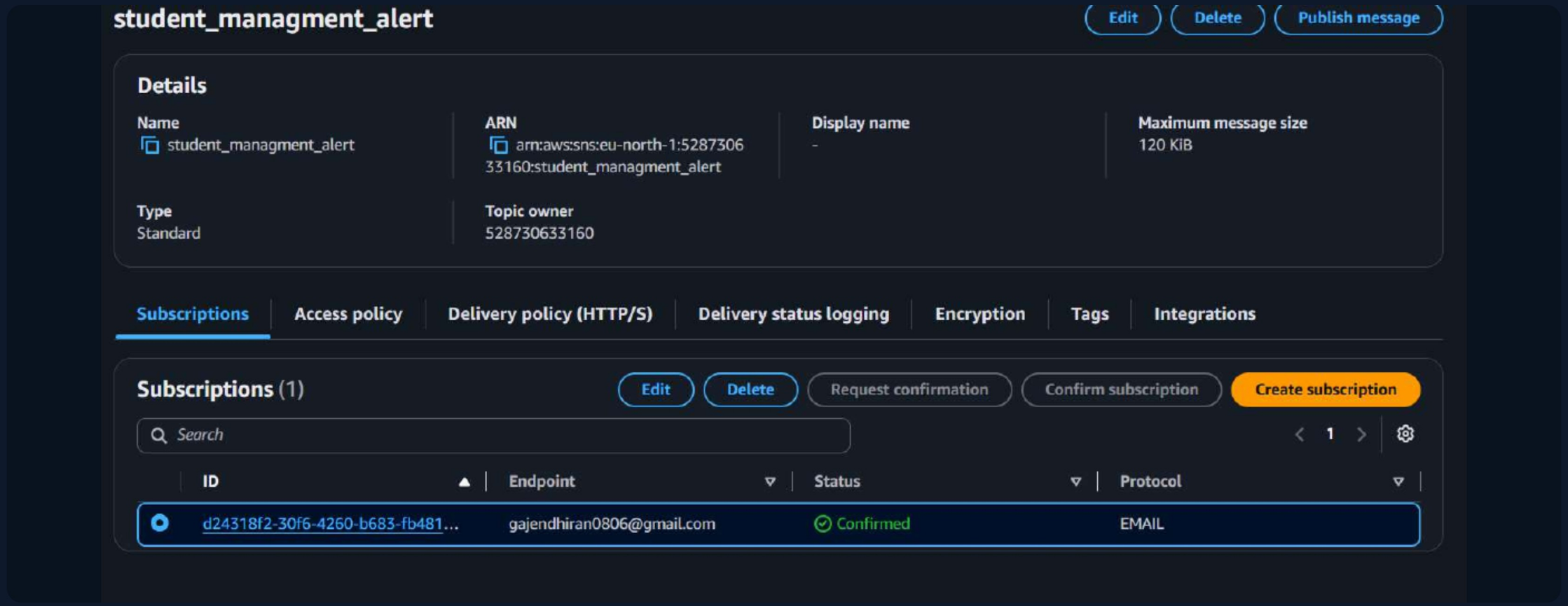
SNS Topic

The student_managment_alert topic



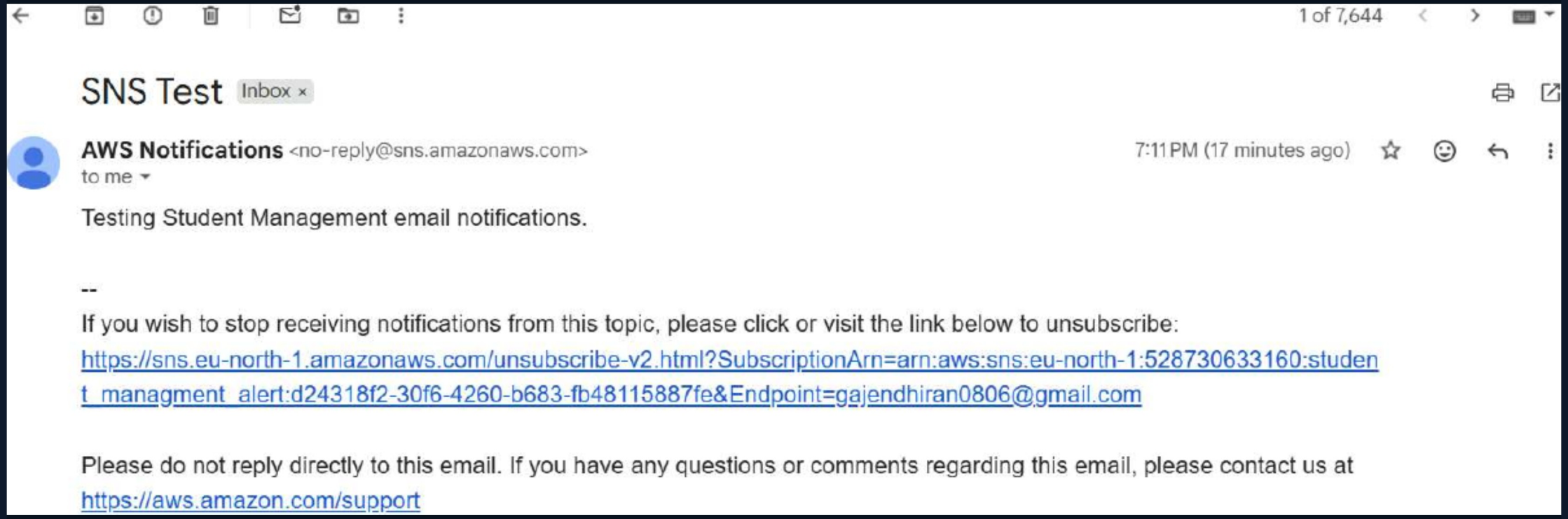
Email Subscription

An email subscription with status Confirmed



Email Notification

Test email received from AWS Notifications



1 of 7,644

SNS Test Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com>
to me

7:11 PM (17 minutes ago)

Testing Student Management email notifications.

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If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:
https://sns.eu-north-1.amazonaws.com/unsubscribe-v2.html?SubscriptionArn=arn:aws:sns:eu-north-1:528730633160:student_managment_alert:d24318f2-30f6-4260-b683-fb48115887fe&Endpoint=gajendhiran0806@gmail.com

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at
https://aws.amazon.com/support

10 · MONITORING

Amazon CloudWatch

CloudWatch collects logs and metrics so we can watch how the services are working.

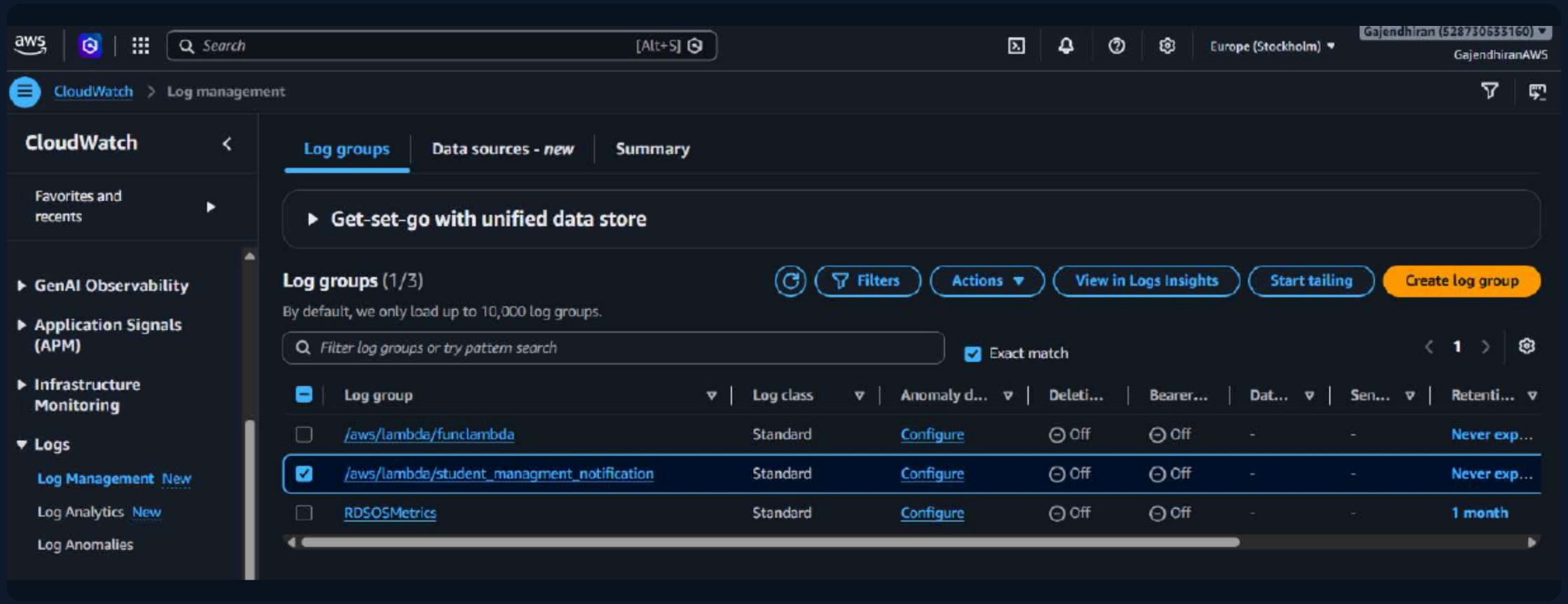
It helps to find errors and check the server load.

In this project

- Log group for the Lambda function
- Log stream for each function run
- EC2 CPUUtilization metric graph

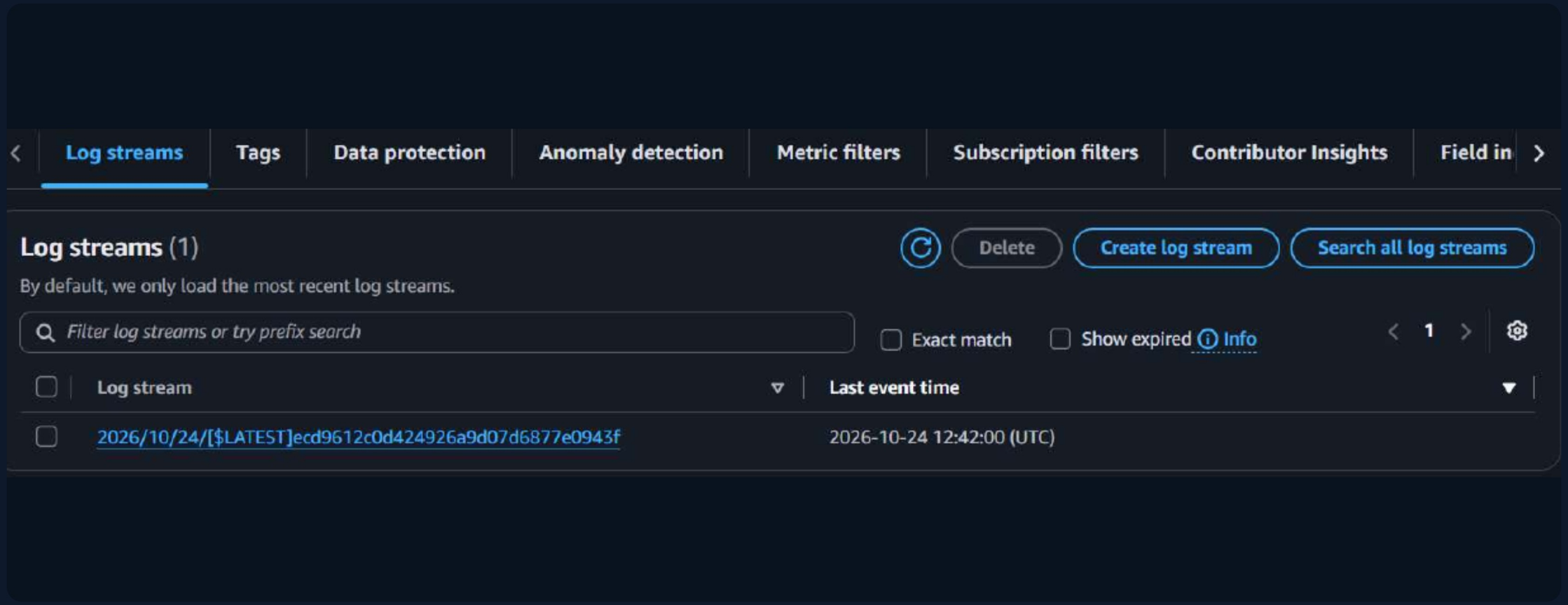
CloudWatch Log Groups

Log group created for the Lambda function



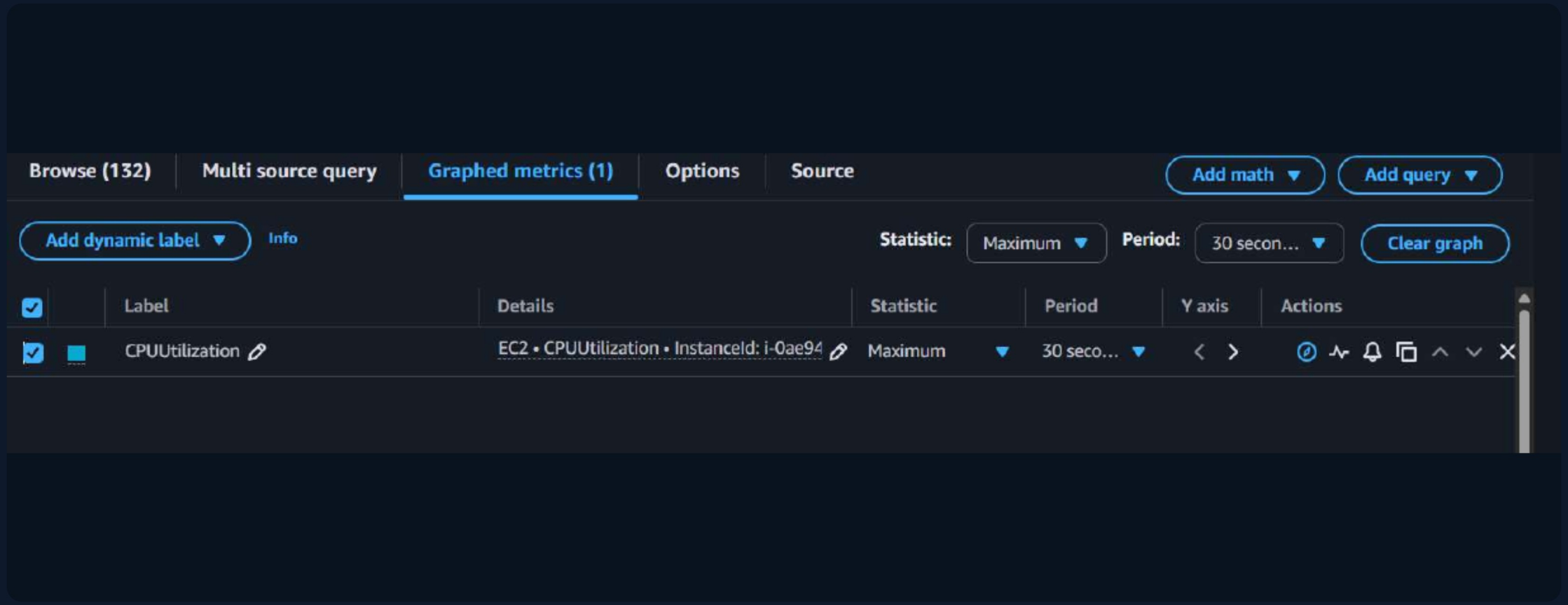
CloudWatch Log Stream

The log stream recorded by the Lambda function



CloudWatch Metrics

EC2 CPUUtilization metric added to the graph



STUDENT MANAGEMENT SYSTEM

Thank You

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